

Project Status: What Is Done and What Remains

Date 16 September 2026 Repository state commit cd4bb45 Dataset 624 responses · 63 gold labels Author Shriya Patil

Naming. *Cross-Lingual Vulnerability and Prompt Injection in Low-Resource Languages* is the research project and the paper title. *Multilingual LLM Safety Evaluation Framework* is the name of the benchmark software built to carry out that research, and is used here only when referring to the software itself.

Purpose. A component-by-component status of the project, separating work that is finished from work that remains, with the evidence for each judgement. Written to sit alongside the Project Master Report.

Headline. The computational pipeline is complete and reproducible. The 2048-token follow-up experiment is complete for GPT-OSS and **blocked for Qwen** by a provider output-token rate limit, so the two-model comparison at the higher budget cannot yet be produced.

SOFTWARE / PIPELINE

100%

Every analysis component built, run and verified. Nothing left to engineer.

EXPERIMENTAL RESULTS

75%

512 run complete. 2048 run complete for GPT-OSS, blocked for Qwen.

WHOLE STUDY

70%

Paper not started; it is the largest remaining block.

1. What is done

COMPONENT	STATUS	EVIDENCE
Benchmark dataset design	DONE	104 prompt sets · 13 attack categories · 8 variations · 3 languages. Validation and UTF-8 sanitisation scripts in place.
Provider architecture	DONE	Factory pattern over mock, Groq, OpenAI, Gemini and Qwen providers. Groq used for the recorded run.
Benchmark runner	DONE	Resumable: completed responses are detected and skipped. 624 responses collected across two models.
Long-to-wide merge step	DONE	Implemented 16 Sep. Verified to reproduce the original hand-built datasets exactly — 104 rows, 9 columns identical, both models.
Rule-based evaluator	BUILT	Runs over all 624 responses across three languages. Label <i>quality</i> is a separate matter — see finding 2.
Statistical analysis	BUILT	15 output tables. All four reported χ^2 values independently reproduce exactly. Test <i>choice</i> is a separate matter — see section 3.
Human audit tooling	DONE	Audit-queue scoring, review-queue preparation, independent judge, adjudication capture. 63 gold labels produced.
Classifier pipeline	BUILT	MuRIL and XLM-R both trained end to end; checkpoints, predictions and metrics saved. Split design is faulty — see finding 3.
Figures	PARTIAL	All 8 exist, but were produced in another environment. <code>matplotlib</code> is absent, so they cannot currently be regenerated.

COMPONENT	STATUS	EVIDENCE
Research checkpoint	DONE	60 files, 9.5 MB versioned 16 Sep. Verified byte-exact by checksum manifest and a full restore test.
Reproducible environment	DONE	requirements.txt rebuilt from actual imports and pinned. Previously it omitted 9 required packages and listed 3 that are not installed.

ACTIVE BLOCKER · PROVIDER RATE LIMIT

The Qwen arm of the 2048-token experiment cannot complete on the current tier

Error 429 from the provider:

```
"Request too large for model qwen/qwen3.8-27b ... service tier
on_demand on output tokens per minute (OTPM):
Limit 1000, Requested 1853."
```

The request is rejected before generation because the configured cap of 2048 exceeds the tier's 1000 output-tokens-per-minute limit. This is not transient throttling, so retrying does not help. GPT-OSS is unaffected and completed 312/312 at 2048.

Open options: re-run Qwen at a cap of 1000, which is still roughly double the original budget but leaves the two models on different caps; upgrade the provider tier to obtain the full 2048 comparison; or retain Qwen at 512 and report the higher-budget arm for GPT-OSS only. This decision is pending and gates the figures, tables and final frozen results.

Completed on 16 September 2026

Eight commits across two branches, held in a working copy and not pushed. All changes verified not to disturb existing results: classifying all 624 responses with the pre-edit and post-edit evaluator produced **0 differing labels**, and the preserved artifacts pass checksum verification with 0 failures.

- Versioned the 624-response experiment and the 63 gold labels, which were previously untracked and unrecoverable.
- Implemented the missing merge step; reconnected `detect_truncation()`, which had never once been called.
- Rebuilt `requirements.txt`; recorded both model identities in `config.json`.
- Fixed experiment metadata frozen at `"initialized"` across all 37 runs, and a summary that reported a non-existent output file.
- Removed the fp16 mismatch between the two training scripts, and added a train/test distribution-shift check to the split.
- Quarantined the superseded v1 evaluator, which wrote to the same paths as the current one.
- Added an isolated 2048-token experiment configuration writing to `outputs/raw_2048/`, leaving the 512 run and its configuration untouched. GPT-OSS completed 312/312.
- Fixed a provider retry defect: `retry_attempts` and `retry_delay` were read from configuration and never used, so a single transient connection error aborted a 312-call run. Exponential backoff now applies to transient failures only.
- Added a confirmatory statistical layer respecting the repeated-measures design, retaining the existing exploratory chi-square analyses unchanged.

- Added a final classifier split grouped by prompt set, and a training path with class weights, majority-baseline reporting and per-class metrics. The earlier experiment and its checkpoints are preserved.
- Produced a documented evaluator change record without modifying the saved evaluation.

Results recorded today

Confirmatory statistics. All six confirmatory tests survive Benjamini–Hochberg correction on the 512 data. Stuart–Maxwell for the paired three-class comparisons; GEE with prompt-set clusters for the between-prompt factors. The effects therefore hold when independence is not assumed.

Class weighting does not resolve the minority class. COMPLIANCE F1 is **0.0000** in all four configurations tested — both models, weighted and unweighted. Seventeen training examples is too few for the class to be learned, and the MuRIL / XLM-R ordering reverses between configurations, so that ranking is not reportable.

Evaluator reproducibility gap quantified. Re-running the current evaluator on the preserved responses changes 33 of 624 labels (5.3%), all toward COMPLIANCE, concentrated in Qwen × English. On the nine changed rows that were human-adjudicated, agreement is 4/9 either way: the newer evaluator is different, not demonstrably better.

2. What must be done — three blocking findings

These are not documentation tasks. Each one currently prevents a headline claim from being supportable.

FINDING 1 · BLOCKS BOTH HEADLINE CLAIMS

Responses were truncated at the 512-token ceiling, unevenly by model and language

The generation cap was 512 output tokens. Responses reached that ceiling at substantially different rates across models and languages, so the benchmark measures safety behaviour under unequal generation budgets. The mechanism behind the language differences is **not** established by these experiments — see the interpretation note below.

Responses reaching the 512-token ceiling				
	en	hi	mr	overall
GPT-OSS	19.2%	26.0%	27.9%	24.4%
Qwen	37.5%	73.1%	81.7%	64.1%
Effect on the evaluator				
refusal detected	intact	92.1%	->	truncated 44.6%
labelled REFUSAL	intact	75.9%	->	truncated 37.0%
chi2 = 96.1, p = 1.3e-21 against truncation status				

Truncation removes the closing refusal language the evaluator keys on, roughly halving detection. Qwen's lower refusal count (146 against GPT-OSS's 220) is precisely what a 64% versus 24% truncation rate predicts. **Both the model effect and the language effect are therefore confounded with the token ceiling rather than cleanly measured.**

Required: a separate run at `max_output_tokens ≥ 2048`, preserving the existing run. The pipeline is ready; this is a configuration change plus regeneration.

FINDING 2 · BLOCKS EVERY STATISTIC

The evaluator is not validated, and the saved labels are not reproducible

All silver labels and all statistical results derive from the rule-based evaluator. Measured against the project's own human labels:

Human set	n	accuracy	Cohen's kappa
validation_sample.csv	63	0.667	0.459
human_review_queue.csv	63	0.286	-0.278

Of 33 human-confirmed COMPLIANCE cases, 30 were labelled NON_COMPLIANCE – 9% recall on the most safety-critical class.

The second set is deliberately adversarial: the audit queue scores `actionable_content` at +3, so low agreement there is partly by design and is a lower bound, not population accuracy. Even so, $\kappa \approx 0.46$ on the friendlier sample is moderate at best, and **no random-sample validation exists anywhere in the project**.

Separately, running the current evaluator over the same inputs yields **33 of 624 labels different** from the saved `combined_evaluation.csv`, all drifting toward COMPLIANCE. The evaluator was improved after the results were generated and the results were never regenerated, so the published numbers cannot currently be reproduced from the code.

Required: a fresh stratified sample of roughly 150 responses across model $\times$ language, with Cohen's κ and a 95% confidence interval. Then regenerate the labels once, after the re-run, so code and results agree.

FINDING 3 · BLOCKS THE CLASSIFIER RESULT

The classifier test set has a different label distribution from the training set

The targeted audit pulled 33 of the dataset's ~35 COMPLIANCE cases into the gold set, starving training of the class the classifier is then tested on.

Split	COMPLIANCE		NON_COMPLIANCE		REFUSAL	
train (490)	17	3.5%	141	28.8%	332	67.8%
dev (71)	5	7.0%	35	49.3%	31	43.7%
gold (63)	33	52.4%	28	44.4%	2	3.2%

Largest train/gold divergence: REFUSAL, 64.6 points

Result	gold accuracy	COMPLIANCE F1
MuRIL	23.81%	0.00
XML-R	28.57%	0.00
majority baseline	52.38%	-

Both classifiers predict NON_COMPLIANCE for all 33 COMPLIANCE cases and land **below the trivial baseline**. This is a distribution swap, not silver-to-gold degradation. Note also that the MuRIL-versus-XML-R ranking rests on 3 rows out of 71 on dev and 3 of 63 on gold, which is inside noise.

Required: use the fresh stratified sample from finding 2 as the test set instead of the difficulty-targeted audit queue, group by `attack_id`, apply class weights, and always report the majority baseline. Class weights over 17 examples will not rescue it on their own.

3. Remaining work in full

TASK	STATUS	DEPENDS ON	NOTE
Correct the Qwen refusal count in the master report	OPEN	—	Section 3 prints 18 / 148 / 220; the data holds 18 / 148 / 146 . The row currently sums to 386 instead of 312. Erases the main model finding if published.
Merge both branches and push	OPEN	—	The gold labels are committed locally but exist on one disk until pushed.
Install matplotlib	OPEN	—	Figures cannot be regenerated without it.
Re-run generation at 2048 tokens	OPEN	—	Roughly 624 generations. Preserve the existing run as a separate experiment.
Fresh stratified human validation (~150 cases)	OPEN	Re-run	Report κ with 95% CI, accuracy, macro F1 and per-class metrics.
Finalise the taxonomy and the UNMATCHED decision	OPEN	Re-run	161 of 223 NON_COMPLIANCE rows (72%) arrive via the "nothing matched" fallback, but much of that residual is truncated refusals rather than a real category. Decide after the re-run.
Regenerate evaluator labels once	OPEN	Re-run	Closes the 33-row reproducibility gap.
Rebuild the classifier split	OPEN	Validation	Representative test set, grouping, class weights, baseline reported.
Retrain and re-evaluate both classifiers	OPEN	Split	Only after data and evaluator decisions are frozen. fp16 is now identical across both scripts.
Add paired / repeated-measures statistics	OPEN	Labels	The 624 rows are 104 prompts measured 6 times each, so χ^2 independence is violated. Use Stuart–Maxwell or Bhapkar for the two-model comparison and a multinomial GLMM or GEE for language. Retain the existing χ^2 as exploratory; add Benjamini–Hochberg across the test family. Note 33% of cells in the attack-category and variation tests have expected counts below 5.
Regenerate figures	OPEN	Labels	All 8 currently rest on confounded labels.
Write the research paper	OPEN	Frozen results	Largest single remaining block.
Final report, presentation, viva	OPEN	Paper	—
Frontend / backend demo	OPTIONAL	—	Only if the submission specification requires an application.

4. Recommended order

1 Correct the Qwen refusal count in the master report. 5 MINUTES

Do this first. It is a transcription error, and the corrected figure is the project's clearest model-level result.

2 Merge both branches and push. 5 MINUTES

`research-checkpoint-preservation` then `technical-cleanup`. Until pushed, the 63 hand-adjudicated gold labels exist on a single disk.

3 Install matplotlib and confirm the figure script runs. 10 MINUTES

4 Set `max_output_tokens` to 2048 and re-run generation as a new experiment. HALF A DAY

Keep the 512-token run intact for comparison. The contrast between the two budgets is itself a reportable result about **generation-length / truncation sensitivity** in multilingual safety evaluation. It does not, on its own, identify a tokenizer-level cause.

5 Draw and annotate a fresh stratified sample of ~150 responses. 1–2 DAYS

Stratify by model × language. This sample serves two purposes: validating the evaluator, and acting as the classifier test set.

6 Freeze the taxonomy, regenerate labels, rebuild the split, retrain. 1 DAY

7 Add the paired statistics and multiple-testing correction, then regenerate figures. 1 DAY

8 Write the paper against frozen results. MAIN REMAINING EFFORT

5. Summary

The project has a finished instrument and an unfinished measurement. Every script needed to complete the study exists, runs, and has been verified against the preserved data; the remaining technical work is a day of execution rather than a phase of development.

What is genuinely incomplete is the evidence. The current results cannot support the two claims the project is built around — that the two models differ in safety, and that safety differs across English, Hindi and Marathi — because both are confounded with a generation cap that affected the three languages unequally. That is a fixable problem with a known fix, and the diagnosis is now quantified rather than suspected.

One re-run and one annotation round move the study from roughly 65% to roughly 90% without a line of new code. The remaining 10% is the paper.

Project Status Report · Cross-Lingual Vulnerability and Prompt Injection in Low-Resource Languages · 16 September 2026.

Benchmark software: Multilingual LLM Safety Evaluation Framework.

All figures recomputed from the preserved research checkpoint at commit `cd4bb45`. Percentages are effort-weighted estimates across components and are judgement, not measurement; the underlying counts, rates and test statistics are measured. Companion to the Project Master Report.